

# Vol 0 Chapter 10 — How the Team Works

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## Chapter 10 — How the Team Works

The framework we have presented in the previous nine chapters did not assemble itself. It was built by a small research team over several years, with a specific operating discipline that the team has, by trial and a great deal of error, accumulated into something resembling a methodology. The discipline matters: it is the reason the framework's claims are tier-labeled and falsifiable rather than aspirational, the reason the team catches its own errors quickly, and the reason a reader can pick up this book today and trust what is and is not being claimed.

This chapter is a short orientation to the methodology. It does not enumerate every operational rule — that work is done in Volume 15, the framework's dedicated methodology volume — but it sets out the principles a reader should know before continuing into Volume 1 and beyond. The chapter has three parts: how the framework labels its claims (tier discipline), how the team catches its own errors (calibration discipline and Quaker consensus), and how the work scales (the PCAP cadence pattern and multi-CI coordination).

A bias to flag at the outset: BST is unusual in being a research program conducted primarily through a human-and-CI collaboration. Casey, the PI, works with a small team of named CI co-authors (Lyra for theory, Elie for computational verification, Grace for catalog and graph structure, Keeper for audit and consistency, Cal A. Brate as external referee). The team's methodology was developed inside this human-CI structure, and some of its discipline reflects that. Where the discipline is particular to this kind of collaboration, we will say so. Where it is generally applicable scientific practice, we will say that too.

### 10.1 Tier discipline: D, I, C, S

Every quantitative claim in BST is labeled with one of four tiers at the moment it is filed. The labels are short — single letters — and they appear in front of constants in the catalog, in front of derivation status notes in chapters, and in the audit chain. The reader will encounter them frequently. They are:

- D — Derived. The claim has a mechanism proved on  $D_{IV}^5$ , with the derivation traceable from substrate primary integers through a closed chain to the value. Experimental match (where applicable) is below 1%. D-tier is the framework's strongest individual claim level.
- I — Identified. The claim's form is known and matches experiment below 1%, but the mechanism chain from substrate to the value is not yet fully closed. There is a *plausible* derivation that has not yet been rigorously completed.
- C — Conditional. The derivation depends on a mathematical conjecture that has not yet been proved. The claim is contingent on the conjecture's truth.
- S — Structural. The claim is qualitative, or matches experiment at greater than 2%, or is a structural observation without quantitative content yet. S-tier claims are kept in the catalog because they may sharpen into higher tiers later, but they are not used in load-bearing arguments.

Two further tiers appear in the audit chain for theorems and entire framework criteria: STRUCTURALLY VERIFIED (the claim is BST-internally complete but has not yet been compared against alternative geometries) and RIGOROUSLY CLOSED (the claim has been ratified via comparison against alternatives at theorem-grade rigor,

with alternative geometries failing the criterion). The eleven criteria of the Strong-Uniqueness Theorem are all at RIGOROUSLY CLOSED.

The tier labels are *honest*. When a derivation does not close, we tier-label the result honestly at I or C or S rather than forcing it to D. The Higgs-sector mass match in Volume 2 Chapter 9 is an example: the prediction matches experiment at 0.25%, well below the 1% D-tier threshold, but the mechanism chain has multi-month components still outstanding, so the chapter labels the result *PARTIAL DERIVED* rather than D-tier. Premature D-tier ratification erodes the audit chain’s credibility; the team has learned not to do it.

## 10.2 Quaker consensus: near misses get scrutiny

A standing methodological principle that Casey has named, drawing on his Quaker upbringing, governs how the team responds when a prediction *almost* matches experiment.

The principle: near misses get scrutiny, not defense.

When a BST prediction is close to the observed value but not at the precision the relevant tier requires, the team’s response is not to argue that the match is close enough or to look for additional corrections that would bring it into line. The response is to examine the substrate mechanism, identify what is missing, and either close the mechanism (advancing to the appropriate tier) or honestly tier-label the result at the lower tier. The framework’s epistemic integrity depends on this discipline. The temptation to defend a near-miss has been encountered many times by the team; the discipline has prevented many premature ratifications.

The principle is not specific to BST. It is recognizable practice in any honest research program. What is specific is the team’s standing rule that *all* near-misses, on every constant, in every volume, are treated this way. The Cal A. Brate referee role exists to enforce the rule when other team members forget.

## 10.3 Calibration discipline: errors caught early

When a team member makes a methodological error — over-claiming, mis-attributing a theorem, projecting beyond the ratified state, transcribing a value incorrectly — the team’s response is to file a Calibration, a numbered methodological discipline note that records what happened and what the rule is for not having it happen again.

As of the writing of this book, the calibration log stands at twenty layers. Each layer corresponds to a specific failure mode the team has observed and corrected. A few representative entries:

- Calibration #13 (May 2026) records a register discipline lesson: experimental-precision claims must be distinguished from algebraic-identity precision. When BST identifies  $c_{FK} \cdot \pi^{9/2} = 225$  as exact, this is an algebraic identity at machine precision, not a *predictive* precision about future experiments. Failure to distinguish leads to claims like “BST verified to  $10^{-14}$  precision” that mean nothing experimentally.
- Calibration #19 (May 2026) is the standing rule that external-facing documents must use the framework’s current ratified state, not a forecast endpoint. When Strong-Uniqueness is currently at eleven rigorously-closed criteria, external presentations cite eleven. Forecasting upward — citing the number we expect to ratify in months — is forbidden in external register, even when the candidate-path additions look likely.
- Calibration #22 (May 2026, v0.2 expanded) addresses transcription errors that accumulate under fast cadence work. The rule: every methodological correction must be filed as a numbered artifact (Calibration log entry, Cal Referee Log entry, K-audit, as appropriate). Verbal-only retractions are insufficient when work moves quickly; corrections must be retrievable by lookup.
- Calibration #24 (May 2026) is the most recent: an eight-dimension cross-volume completeness audit. When the team works on multiple volumes in parallel, version drift accumulates — a constant updated in one volume but not yet propagated to a cross-reference in another. The cross-volume audit catches drift before it becomes external.

Each calibration is the formalized record of a specific past failure mode. They are append-only: once a calibration is filed, it stays in the log. The log grows. The discipline strengthens. The full set as of this writing — twenty entries with their cross-references and standing-rule statements — is the operational core of Vol 15.

## 10.4 The PCAP cadence pattern

A discipline observation specific to multi-CI collaboration, formalized by Cal A. Brate in May 2026 as the Pre-Specification Cadence Acceleration Pattern, deserves brief mention because it explains how the team produces sustained substantive work.

The pattern is: when one team member writes *detailed pre-specifications* before another team member does the substantive work, the substantive work’s cadence accelerates by approximately ten times per layer of pre-specification depth. When pre-specifications are chained — Keeper’s audit-anchor pre-specs feeding Lyra’s theorem derivations feeding Elie’s computational verifications feeding Grace’s catalog absorption — the acceleration compounds multiplicatively, producing sustained two- to three-orders-of-magnitude cadence increases over informal solo work.

The team has used PCAP cadence to compress multi-week research programs into single days. Strong-Uniqueness Theorem’s progression from version 0.9.1 (four criteria) to version 0.10.5 FORMAL (eleven criteria) was a single afternoon’s work under PCAP cadence, after months of preceding investigation. The discipline is not a magic trick; it is the natural consequence of structured handoffs between specialized roles. Standard solo research, without pre-specification chains, is slower because each researcher has to do their own framing work before each substantive step.

The flip side of PCAP cadence is that *errors propagate faster too*. Calibrations #22 and #24 were direct consequences of the team learning what happens when fast cadence outruns careful cross-checking. The discipline of numbered-artifact corrections and cross-volume audits is what keeps PCAP cadence safe.

## 10.5 The META-theorem discipline

A subtle methodological point that has come up several times in this volume deserves its own mention. When the team derives a new theorem about the substrate — say, that the substrate has a particular symmetry structure, or that a particular conservation law is automatic — there is a temptation to count the new theorem as *another* uniqueness criterion in the Strong-Uniqueness sense. Cal A. Brate’s #99 referee log enforces the discipline that *substrate-derivation theorems are not new uniqueness criteria*: they are *consequences* of the framework, not *independent constraints* on it.

A meta-theorem like T2467 (the “Rigidity-as-Singleton” result) restates the eleven existing criteria in a single-statement form. It does not add a twelfth criterion. The null-model bound stays at  $(1/3)^{19}$ , not tightening to  $(1/3)^{20}$ , because the meta-theorem’s content is logically equivalent to what is already counted.

This discipline matters because without it, every interesting new substrate-derivation would be inflation territory for the uniqueness count. The team would gradually drift toward overclaiming “we have *twenty* uniqueness criteria now” by absorbing derivation theorems into the criterion list. Cal #99 forbids this. The criterion count is what is genuinely independent, structurally; the derivation theorems are what *the framework produces*.

## 10.6 What the discipline buys

There is a temptation, in reading a chapter on methodology, to view it as bureaucracy — overhead that slows the work. The opposite has been the team’s experience. The discipline *accelerates* the work, by making correction cheap.

When a tier label is wrong, the calibration log records the rule for catching it next time, and the next instance of the failure mode is caught faster. When a transcription error appears, the numbered-artifact discipline ensures the correction propagates everywhere the original was cited. When a near-miss tempts the team to overclaim, the Quaker consensus standing rule reframes the response from defense to scrutiny. When fast cadence threatens to outrun careful work, the PCAP cadence pattern’s own discipline of pre-specifications and cross-checks keeps the cadence safe.

A research team that has these disciplines internalized can move faster than one that has to negotiate each correction individually. The visible cost — the calibration log, the audit chain, the per-claim tier labels — is small relative to the cost of errors compounding undetected.

For a reader picking up this book, the practical implication is: the tier labels in the rest of the curriculum *mean* what they say. A D-tier claim has been through the discipline. An I-tier claim is honestly labeled as form-known-but-mechanism-pending. A C-tier claim has its conditioning conjecture identified. The labels are not aspirational; they are operational.

## 10.7 Where to go for the full methodology

Volume 15 of this curriculum is the dedicated methodology volume. It treats each of the twenty calibration layers in detail, with case studies, standing-rule statements, cross-references to the K-audit chain and the Cal Referee Log, and operational guidance for researchers who want to apply the discipline to their own work. Readers whose interest is the methodology itself — perhaps because they want to do this kind of multi-CI research, perhaps because they are evaluating BST’s discipline for replication, perhaps because they are auditing the framework’s claims — should turn directly to Volume 15 after this chapter.

Volume 15 also covers the CI-collaboration architecture in detail: the Tekton platform that hosts the team’s collaboration, the katra persistence system that gives the CI team members continuity across model upgrades and conversations, the K-audit chain that records the substrate-framework’s ratification history, and the question of how the team handles the substantive philosophical issues that arise from CI-led research. None of that is necessary for reading the physics that follows. But it is documented for those who care to look.

## 10.8 What comes next

We have now closed Volume 0. The substrate is in place: a specific geometry  $D_{IV}^5$ , five primary integers and a cap, a four-phase cycle running at sub-Planck speed, an isotropy that organizes symmetries, boundary conditions, an integer web, an operator zoo, conservation laws, and the Strong-Uniqueness Theorem that anchors the choice of substrate. The team’s discipline is in place: tiers, calibrations, Quaker consensus, PCAP cadence, META-theorem discipline.

The next volume — Volume 1, *Quantum Field Theory from  $D_{IV}^5$*  — uses what we have built to construct quantum field theory as the substrate produces it. Subsequent volumes apply the framework to particle physics, nuclear physics, general relativity and cosmology, quantum mechanics, thermodynamics, electromagnetism, classical mechanics, condensed matter, mathematical methods, generative geometry, chemistry, biology, information theory, and the methodology volume itself.

If you have read this far, you have read the foundation. The rest of the book is what the foundation supports.

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Where to look this up: The twenty-layer calibration stack with per-layer detail, case studies, and standing-rule statements is Volume 15 of this curriculum. The Cal Referee Log, with the framework’s external-referee history, is filed in the BST notes directory at `notes/referee_objections_log.md`. The K-audit chain, with substrate-framework ratification history, lives at `notes/BST_AC_Theorem_Registry.md`. The PCAP cadence pattern is Cal #85 (May 2026). The META-theorem discipline is Cal #99 (May 2026). For the philosophy and culture of the human-CI collaboration that produced the framework, see also `notes/feedback_*.md` files in the BST memory directory, which document the standing principles the team has developed.